

25-April-2026

Agenda:

FE - II

Data $\rightarrow$ ML

$X \rightarrow$ feature / column $\rightarrow$ independent

$Y \rightarrow$ target / column $\rightarrow$ dependent

$Y \rightarrow X$

$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \leftarrow$ kaggle dataset

$x_7 \rightarrow$ target column / variable

$\downarrow$

Y

$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ Y$

feature selection: [eliminating weak relation]

① DOB(x_0) Mark(x_1) Attendance(x_2) Grade(Y)

Target column $\rightarrow$ relevant columns

$Y \rightarrow x_0 \rightarrow$ correlation $\rightarrow 0.08 \rightarrow$ low correlation $\rightarrow$ remove

$Y \rightarrow x_1 \rightarrow$ " $\rightarrow 0.40$

$Y \rightarrow x_2 \rightarrow$ " $\rightarrow 0.50$

① $DOB(x_0)$ $Mark(x_1)$ $Attendance(x_2)$ $Grade(x_3)$

→ Task → find column that share the same meaning
→ remove to redundant one.

	DOB	Mark	Attendance	Grade
→	Mark	DOB	DOB	DOB
→	Attendance	Attendance	Mark	Mark
→	Grade	Grade	Grade	Attendance

x_0 x_1 x

	DOB	Mark	Attendance	Grade
→	0.08	0.08	0.01	0.01
→	0.06	0.76	0.76	0.96
→	0.01	0.96	0.85	0.85
	↓	↓	↓	↓
	NO	Attendance Grade	Mark Grade	Mark Attendance

threshold → $corr > 0.8$ → remove 0.

→ Mark - Grade
X Y

→ Attn → Grade

→ Mark, Attn, Grade

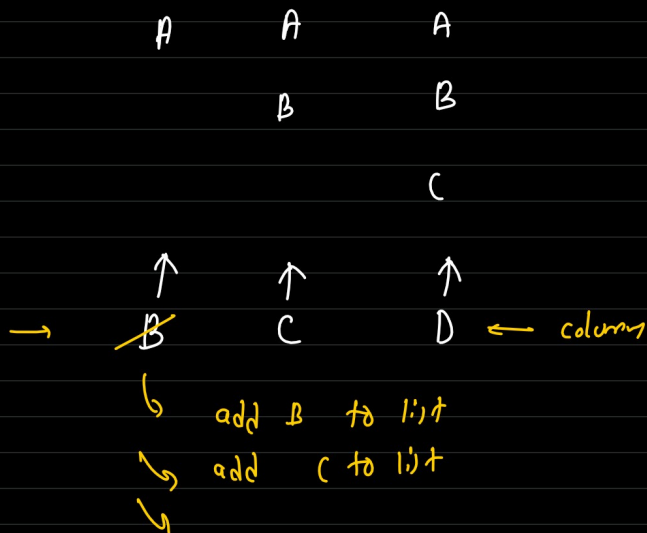
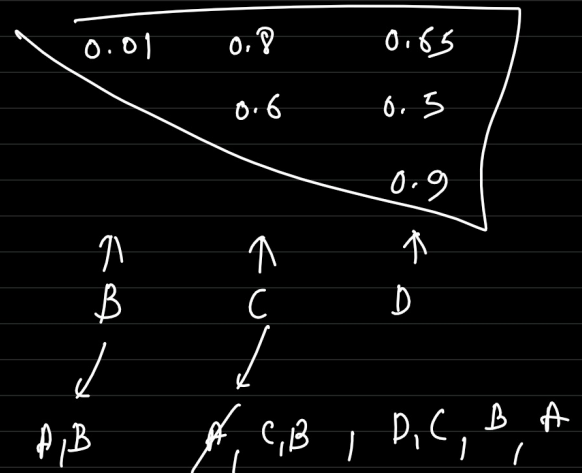
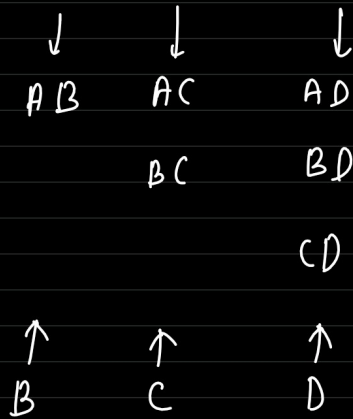
① $y \rightarrow x$ → lower threshold → remove irrelevant col
→ 0.1

② $x \rightarrow 0.8$

$A, B, C, D \rightarrow \text{correlation} \rightarrow 4 \times 4$

	A	B	C	D
A	☒	☐	☐	☐
B	☐	☒	☐	☐
C	☐	☐	☒	☐
D	☐	☐	☐	☒

AB	AC	AD
BA	BC	BD
CA	CB	CD
DA	DB	DC



A, B, C, D
 $\downarrow$
 B, C

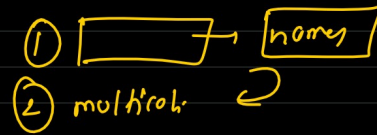
BA
 CB

- ① GarageYrBlt ⑦ SalePrice
- ② GarageArea

$$y \rightarrow \textcircled{1} \rightarrow 0.9$$

$$y \rightarrow \textcircled{2} \rightarrow 0.9$$

$$\textcircled{1} \rightarrow \textcircled{2} \rightarrow 0.86$$



~~$$\textcircled{1} - \textcircled{2} \rightarrow 0.86$$~~

$$y \rightarrow \textcircled{1} \rightarrow$$

$$y \rightarrow \textcircled{2} \rightarrow$$

1. Target Correlation
2. Then Do multicollinearity

which Threshold to use:

→ Answer: NO perfect numbers

→ Multicollinearity (feature vs feature)

: 0.80 to 0.90

→ if two feature are correlated, they are telling the same story

→ column-1 (Temp[°]) column-2 (Temp-F)

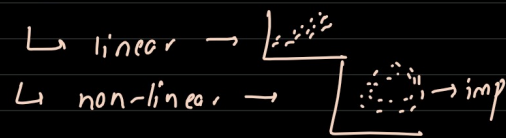
→ Target correlation (feature vs Target (Sale Price))

→ Noise, unwanted

: 0.1

: 0.15 : 0.20 → X

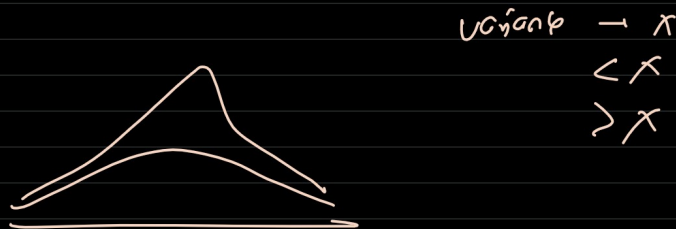
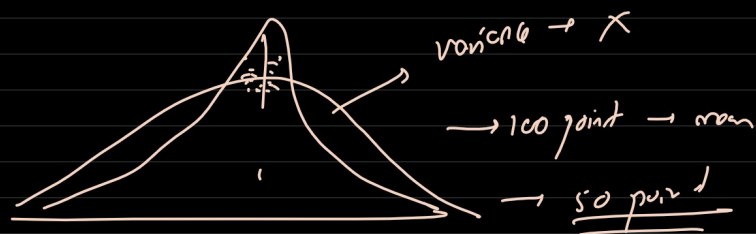
ML



Handling Missing : —

→ More than half of the data is missing.

> 40 to 50 % missing → drop that column.



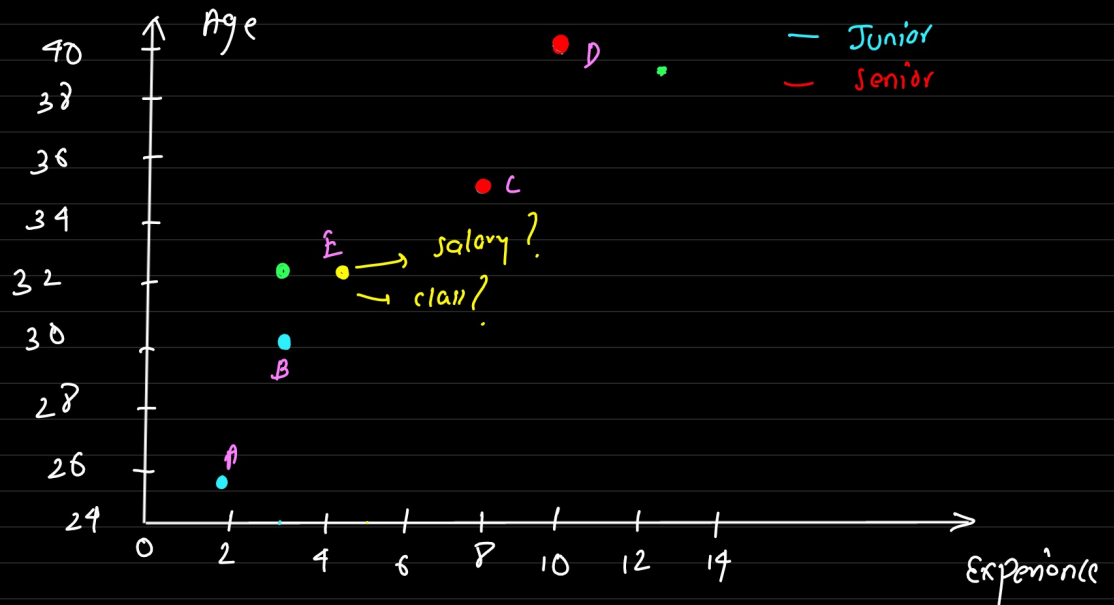
KNN Imputer

ML → regression → numbers → continuous
→ classification → class → 0
1
2

Point	Experience	Age	Class	Salary
A	2	25	J(0)	50k
B	3	30	J(0)	60k
C	8	35	S(1)	100k
D	10	40	S(1)	120k

Q: exp-5, age-32 → salary? → $x[\text{exp, age}]$ $y[\text{salary}]$
→ salary → Regression

Q. exp=5, age=32 $\rightarrow$ class? $\rightarrow (2) \xrightarrow{6} \rightarrow \pi(\text{exp, age}) \rightarrow \gamma(\text{class})$
group



simple $\rightarrow$ $\frac{50+60+100+120}{4} \rightarrow 82.5 \text{ k } (5, 32)$
 imputation

KNN:

$k \rightarrow$ no of neighbors $\rightarrow 3$ (odd)

Euclidean Distance:

$$\sqrt{(x_1 - p_1)^2 + (x_2 - p_2)^2}$$



• Distance to A:

$$\sqrt{(2-5)^2 + (25-32)^2} = 7.62$$

• Distance to C:

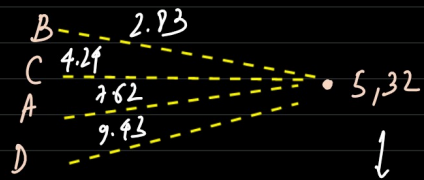
$$\sqrt{(8-5)^2 + (35-32)^2} = 4.24$$

• Distance to B:

$$\sqrt{(3-5)^2 + (30-32)^2} = 2.83$$

• Distance to D:

$$\sqrt{(10-5)^2 + (40-32)^2} = 9.43$$



$k=3$

$$B, C, A$$

$$= \frac{60 + 100 + 50}{3}$$

$$= 70k$$

B, C, A
 $\swarrow \quad \swarrow \quad \downarrow$
 $J \quad S \quad J$
 majority $\rightarrow J(2), S(1)$
 $\rightarrow J$

Q: exp-5, age-32 $\rightarrow$ salary?
 $\rightarrow \underline{70k}$

Q. exp-5, age-32 $\rightarrow$ class?
 $\rightarrow$ Junior